The "Taylor step" in the proof of Proposition 2.4 (Slater $\Rightarrow$ CQ)

Optimization — Semana 5

Setting. As in Proposition 2.4, the feasible set is defined by inequality constraints only,

$$C = \{x \in \mathbb{R}^n : g_i(x) \le 0,\ i = 1, \dots, p\}, \qquad I(\overline{x}) = \{i : g_i(\overline{x}) = 0\},$$

with each g_i of class $\mathcal{C}^1$. The claim to justify is:

> For each $\epsilon > 0$ there exists $t_\epsilon > 0$ such that $\overline{x} + t d_\epsilon \in C$ for all $t \in (0, t_\epsilon]$,

where $d_\epsilon = d_0 + \epsilon d$ satisfies $\langle \nabla g_i(\overline{x}), d_\epsilon \rangle < 0$ for all $i \in I(\overline{x})$ (this was established just before in the proof). The claim is the case $v = d_\epsilon$ of the following lemma. Convexity plays no role here; it was used only to obtain the strict inequality.

Lemma 1. *Let $\overline{x} \in C$, and let $v \in \mathbb{R}^n$ satisfy*

$$\langle \nabla g_i(\overline{x}), v \rangle < 0 \qquad \forall i \in I(\overline{x}). \tag{1}$$

Then there exists $\tau > 0$ such that $\overline{x} + tv \in C$ for every $t \in (0, \tau]$.

Proof. We will show that for each index $i \in \{1, \dots, p\}$ there exists a number $\tau_i > 0$ such that

$$g_i(\overline{x} + tv) < 0 \qquad \forall t \in (0, \tau_i]. \tag{2}$$

Step 1 constructs such a τ_i for the active indices $i \in I(\overline{x})$, and Step 2 for the inactive ones $i \notin I(\overline{x})$. Only once every τ_i has been obtained, Step 3 sets $\tau := \min_{1 \le i \le p} \tau_i$ and checks that it has the required property.

Step 1: active constraints, $i \in I(\overline{x})$. Write $\alpha_i := \langle \nabla g_i(\overline{x}), v \rangle$, so $\alpha_i < 0$ by (1). Since g_i is of class $\mathcal{C}^1$, it is differentiable at $\overline{x}$, and by the first-order Taylor theorem

$$g_i(\overline{x} + tv) = g_i(\overline{x}) + t\, \langle \nabla g_i(\overline{x}), v \rangle + r_i(t), \qquad \lim_{t \to 0^+} \frac{r_i(t)}{t} = 0.$$

Because i is active, $g_i(\overline{x}) = 0$, hence for $t > 0$

$$\frac{g_i(\overline{x} + tv)}{t} = \alpha_i + \frac{r_i(t)}{t}. \tag{3}$$

Apply the definition of the limit $r_i(t)/t \to 0$ with tolerance $-\alpha_i/2 > 0$: there exists $\delta_i > 0$ such that $|r_i(t)/t| < -\alpha_i/2$ for all $t \in (0, \delta_i)$. Setting $\tau_i := \delta_i/2$, so that $(0, \tau_i] \subset (0, \delta_i)$,

$$\left| \frac{r_i(t)}{t} \right| < -\frac{\alpha_i}{2} \qquad \forall t \in (0, \tau_i].$$

Substituting in (3), for every $t \in (0, \tau_i]$,

$$\frac{g_i(\overline{x} + tv)}{t} < \alpha_i - \frac{\alpha_i}{2} = \frac{\alpha_i}{2} < 0,$$

and multiplying by $t > 0$ gives $g_i(\overline{x} + tv) < \frac{\alpha_i}{2} t < 0$. Hence the number $\tau_i = \delta_i/2$ satisfies (2), which proves its existence for every active index.

Step 2: inactive constraints, $i \notin I(\overline{x})$. Since $\overline{x} \in C$ and i is not active, $\beta_i := g_i(\overline{x}) < 0$. The function g_i is continuous at $\overline{x}$ (it is $\mathcal{C}^1$), so for the tolerance $-\beta_i > 0$ there is $\delta_i > 0$ with

$$\|y - \overline{x}\| < \delta_i \implies |g_i(y) - g_i(\overline{x})| < -\beta_i \implies g_i(y) < \beta_i - \beta_i = 0.$$

Put $\tau_i := \dfrac{\delta_i}{\|v\| + 1} > 0$. For $t \in (0, \tau_i]$,

$$\|(\overline{x} + tv) - \overline{x}\| = t\|v\| \leq \frac{\delta_i \|v\|}{\|v\| + 1} < \delta_i,$$

hence $g_i(\overline{x}+tv) < 0$. Thus $\tau_i = \delta_i/(\|v\|+1)$ satisfies (2), which proves its existence for every inactive index. (The $+1$ only avoids dividing by zero when $v = 0$, which can happen if $I(\overline{x}) = \emptyset$.)

Step 3: conclusion. Every index $i \in \{1, \ldots, p\}$ is either active or inactive, so Steps 1–2 provide a number $\tau_i > 0$ satisfying (2) for each i. There are finitely many of them, so $\tau := \min\{\tau_1, \ldots, \tau_p\} > 0$. For $t \in (0, \tau]$ we have $t \leq \tau_i$ for every i, so by Steps 1–2, $g_i(\overline{x} + tv) < 0$ for all $i = 1, \ldots, p$, i.e. $\overline{x} + tv \in C$. □

Corollary. *Under the hypothesis of Lemma 1, $v \in T(\overline{x}, C)$. In particular $d_\epsilon \in T(\overline{x}, C)$ for every $\epsilon > 0$.*

Proof. Let τ be given by Lemma 1 and set $t_k := \tau/k$ and $d_k := v$ for $k \in \mathbb{N}^*$. Then $t_k \to 0^+$, $d_k \to v$ (constant sequence), and $t_k \in (0, \tau]$, so $\overline{x} + t_k d_k = \overline{x} + t_k v \in C$ for all k. By Definition 2.1, $v \in T(\overline{x}, C)$. Taking $v = d_\epsilon$, which satisfies (1), gives $d_\epsilon \in T(\overline{x}, C)$ with $t_\epsilon := \tau$. □

This is exactly the highlighted sentence, and the rest of the proof of Proposition 2.4 follows: $d_\epsilon \to d_0$ as $\epsilon \to 0^+$, and $T(\overline{x}, C)$ is closed (Proposition 2.1), so $d_0 \in T(\overline{x}, C)$.

Why each ingredient is needed

Remark (The inequality must be strict). Step 1 only works because $\alpha_i < 0$: the Taylor remainder $r_i(t) = o(t)$ has unknown sign, and a strictly negative linear term is what dominates it for small t. If $\alpha_i = 0$, the sign of $g_i(\overline{x} + tv)$ is decided by $r_i(t)$ and nothing can be concluded. This is why the proof does not apply the argument to $d_0 \in L(\overline{x}, C)$ directly (which only gives ≤ 0), but to the perturbed d_ϵ, and recovers d_0 through closedness of the tangent cone.

Example. Take the example of the notes: $g_1(x) = x_1^2 + x_2$, $g_2(x) = x_1^2 - x_2$, $\overline{x} = (0,0)$, so $C = \{(0,0)\}$. The direction $d_0 = (1,0)$ lies in $L(\overline{x}, C)$ with $\langle \nabla g_1(\overline{x}), d_0 \rangle = \langle \nabla g_2(\overline{x}), d_0 \rangle = 0$, yet

$$g_1(\overline{x} + td_0) = t^2 > 0 \qquad \forall t > 0,$$

so $\overline{x} + td_0 \notin C$ for every $t > 0$: the conclusion of Lemma 1 fails with non-strict inequalities. Consistently, Slater fails here: $g_1(x) + g_2(x) = 2x_1^2 \geq 0$, so no $\hat{x}$ has $g_1(\hat{x}) < 0$ and $g_2(\hat{x}) < 0$, and no direction d with both $\langle \nabla g_i(\overline{x}), d \rangle < 0$ exists (they would require $d_2 < 0$ and $-d_2 < 0$).

Remark (Inactive constraints). For $i \notin I(\overline{x})$ no condition on v is needed: continuity alone keeps $g_i < 0$ near $\overline{x}$. The argument uses that there are finitely many constraints, so the minimum of the τ_i is positive.

Remark (No equality constraints). The lemma uses that C is described by inequalities only. With a constraint $h(x) = 0$, the condition $\langle \nabla h(\overline{x}), v \rangle = 0$ does not give $h(\overline{x} + tv) = 0$ unless h is affine (e.g. $h(x) = x_1^2 + x_2^2 - 1$, $\overline{x} = (1, 0)$, $v = (0, 1)$: $h(\overline{x} + tv) = t^2 \neq 0$). This is why Theorem 2.2 (MFCQ) replaces the ray $\overline{x} + tv$ by a curve $\gamma_k(t)$ from the implicit function theorem.